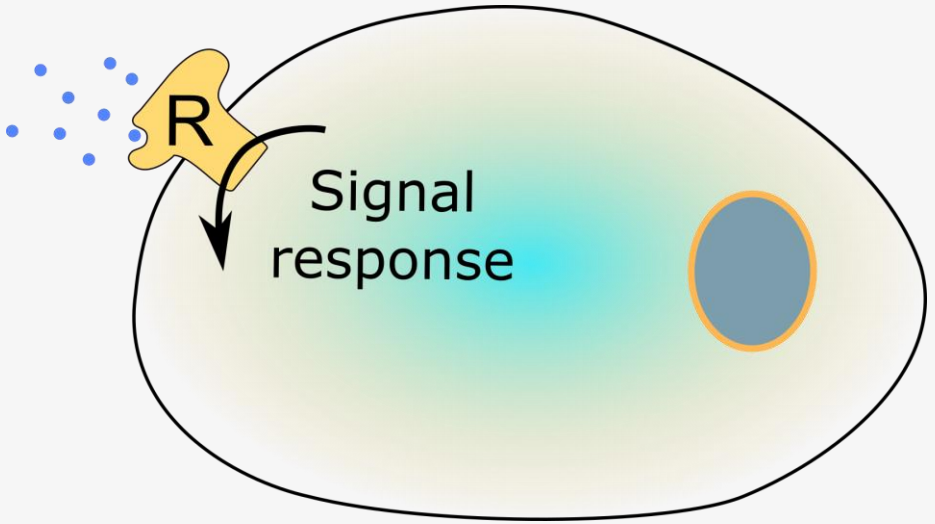
PHAR0002: Partial agonists, inverse agonists, spare receptors etc.

Dr. Guy Moss

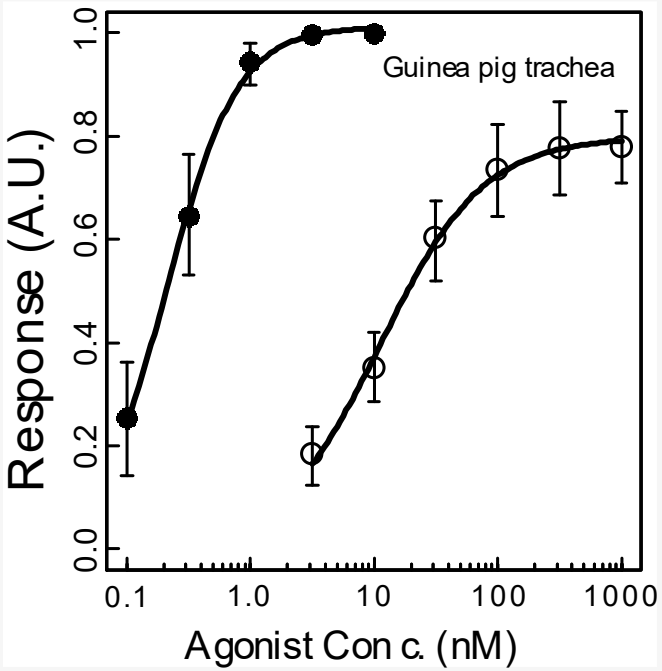


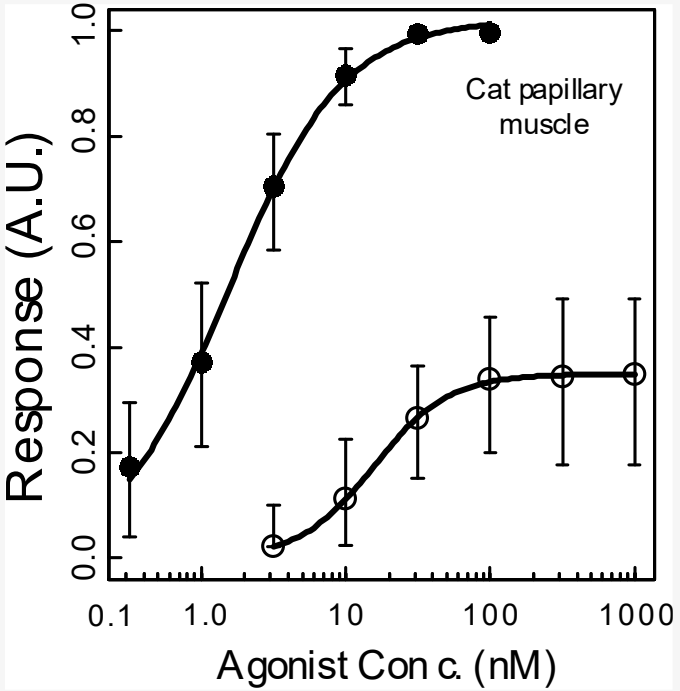
Some questions to think about…

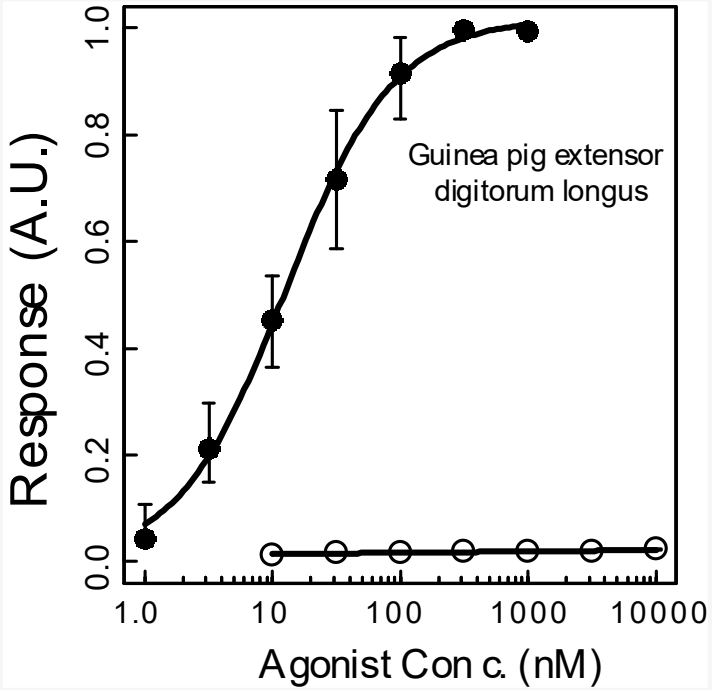
1. If a drug were to bind irreversibly, could you measure the affinity (K_B value)?

2. Why might the Hill coefficient be different from one?

Some unexplained observations…



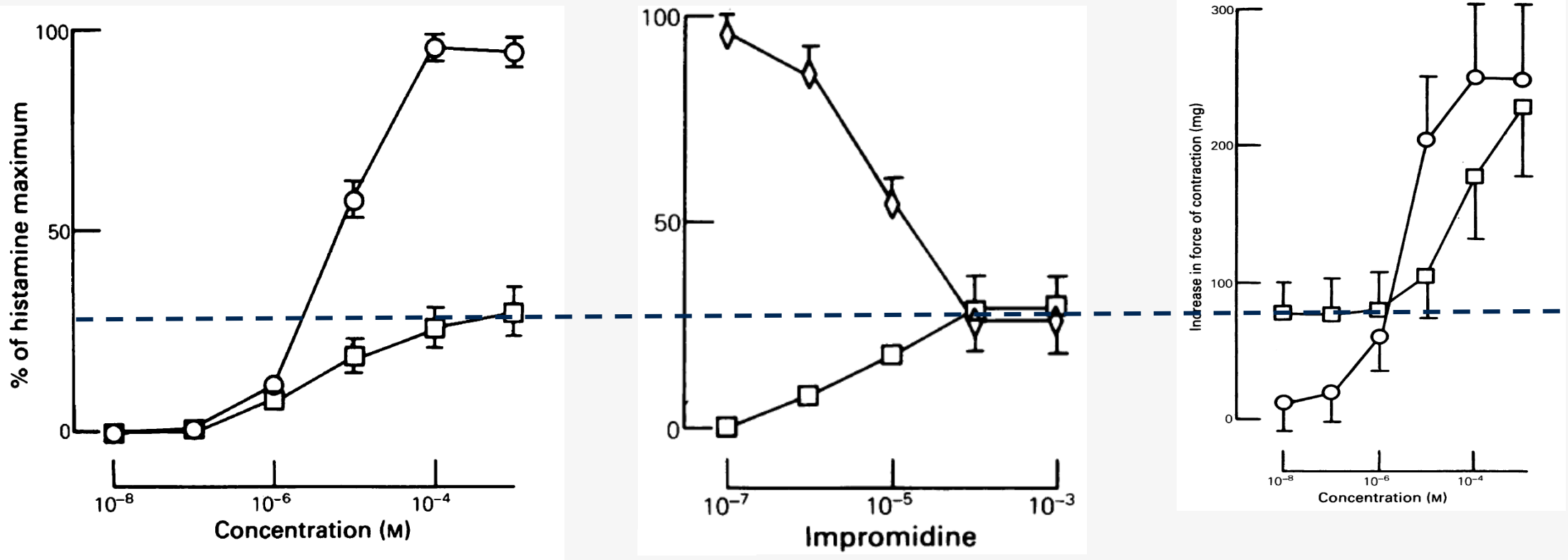




● Isoproteranol
○ Prenalterol

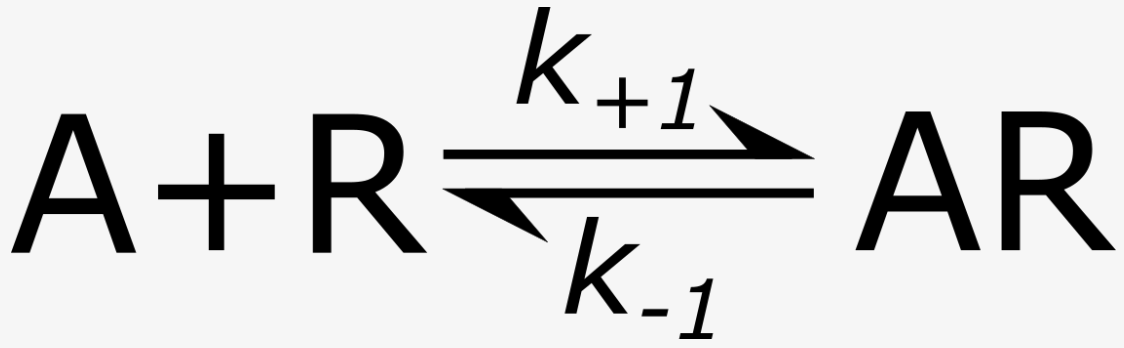
Data reproduced from Kenakin and Beek (1980).

More observations…



Modified from T.A.H. English *et al*., (1986), *Br. J. Pharmacol.*, **89**, 335 – 340.

How can we explain this?

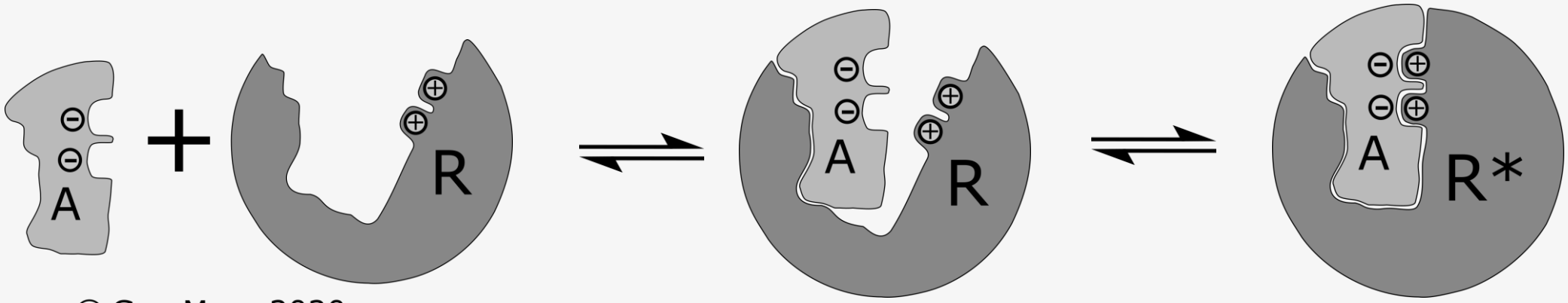


But why are receptors activated by agonists?...

The del Castillo-Katz Mechanism…



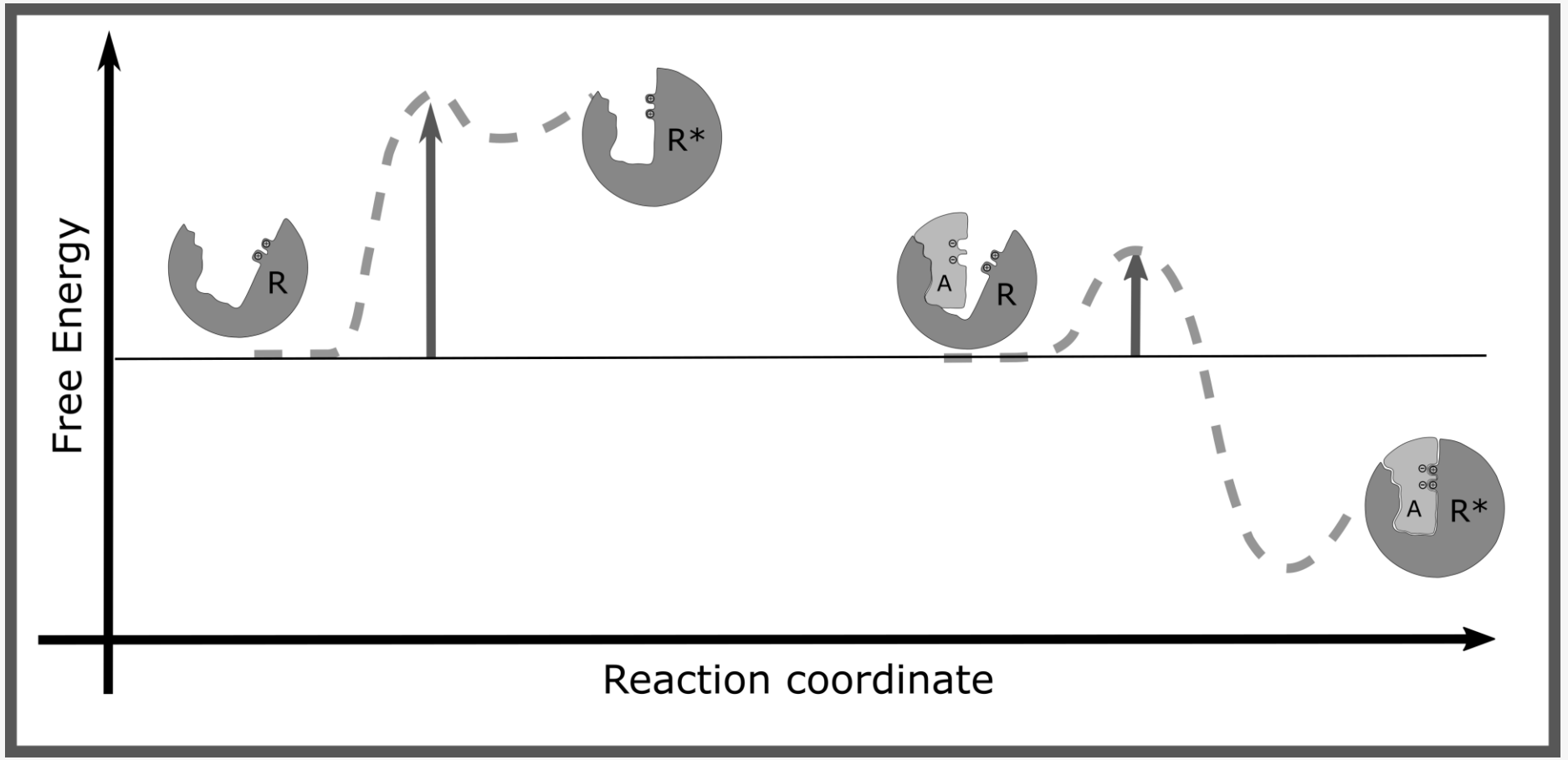
Bonds are the answer…



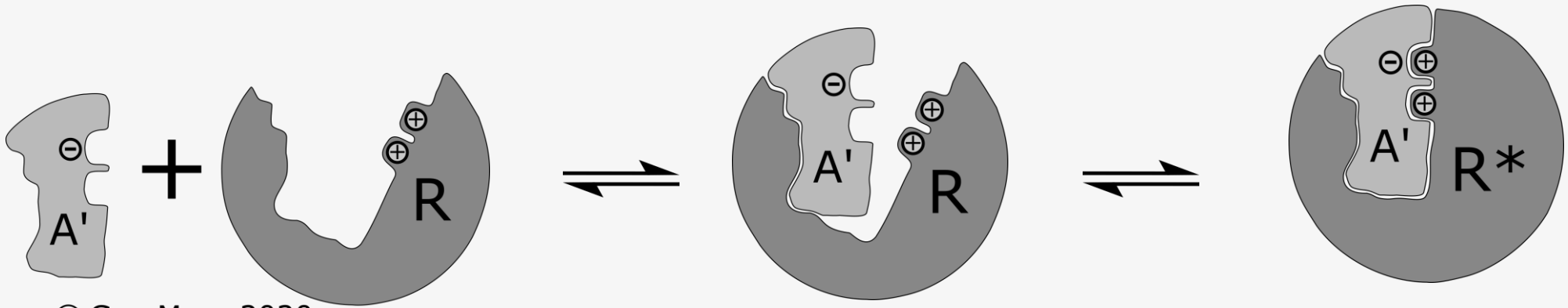
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Warning: In what follows we are using charges as an aid to understanding the concepts of receptor-agonist interactions. **Do not take this literally** – receptor activation can occur via any bond type and charge interactions such as this are not meant to be a realistic representation of all (or indeed any) specific receptor activation process.

In terms of the energetic landscape…

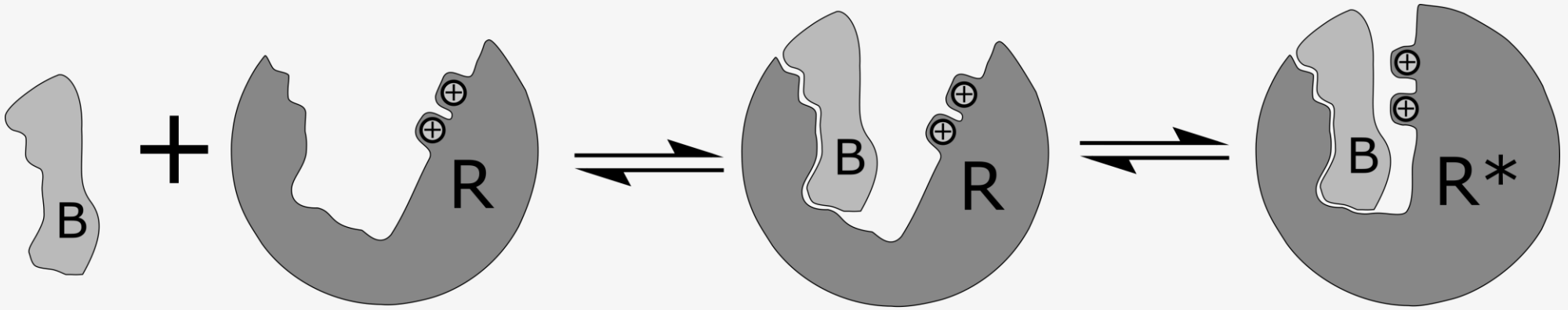


What happens if we change the agonist?



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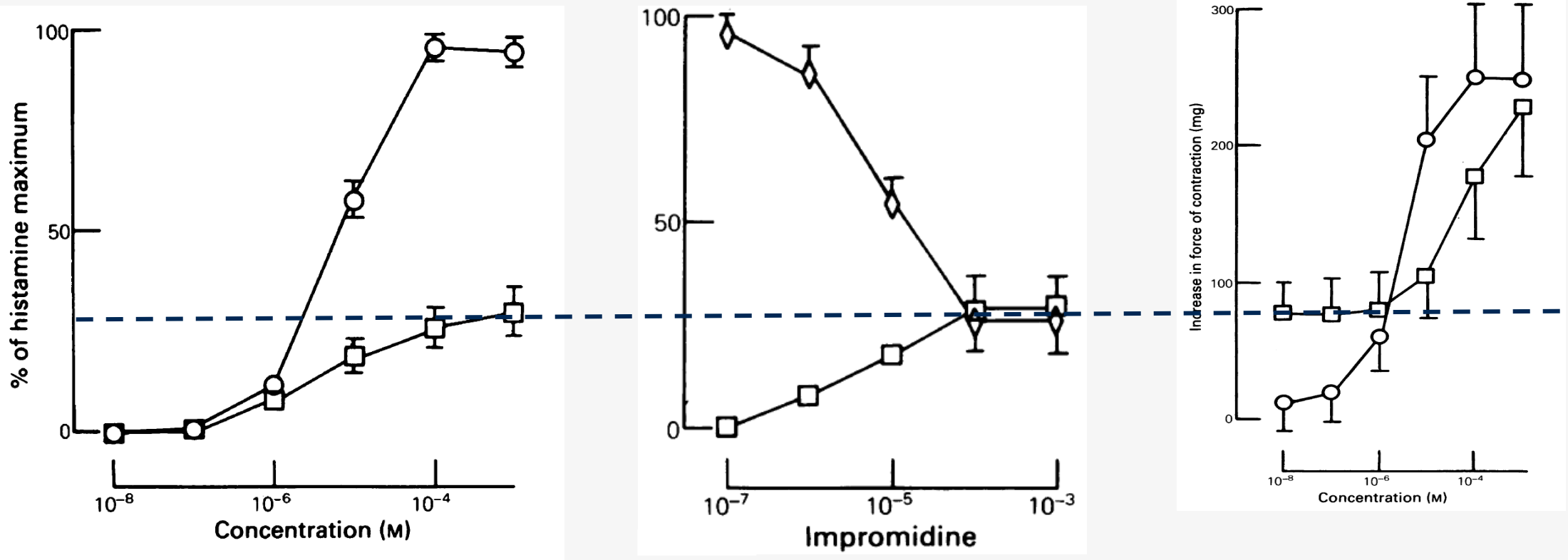
What if we go further with our changes?



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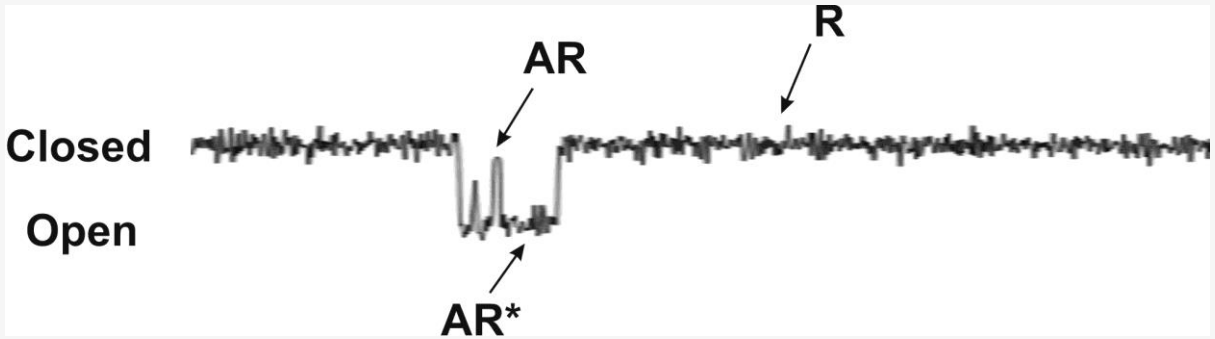
Reversible competitive antagonist

Back to the observations…



Modified from T.A.H. English *et al*., (1986), *Br. J. Pharmacol.*, **89**, 335 – 340.

We know this is right for ligand-gated ion channels - because we can see it happening…



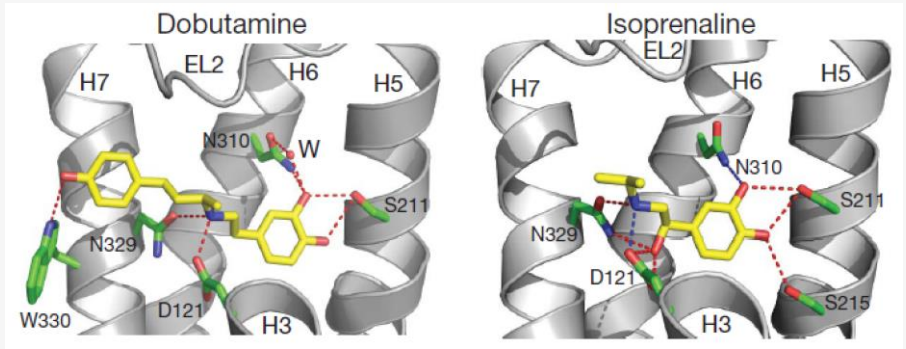
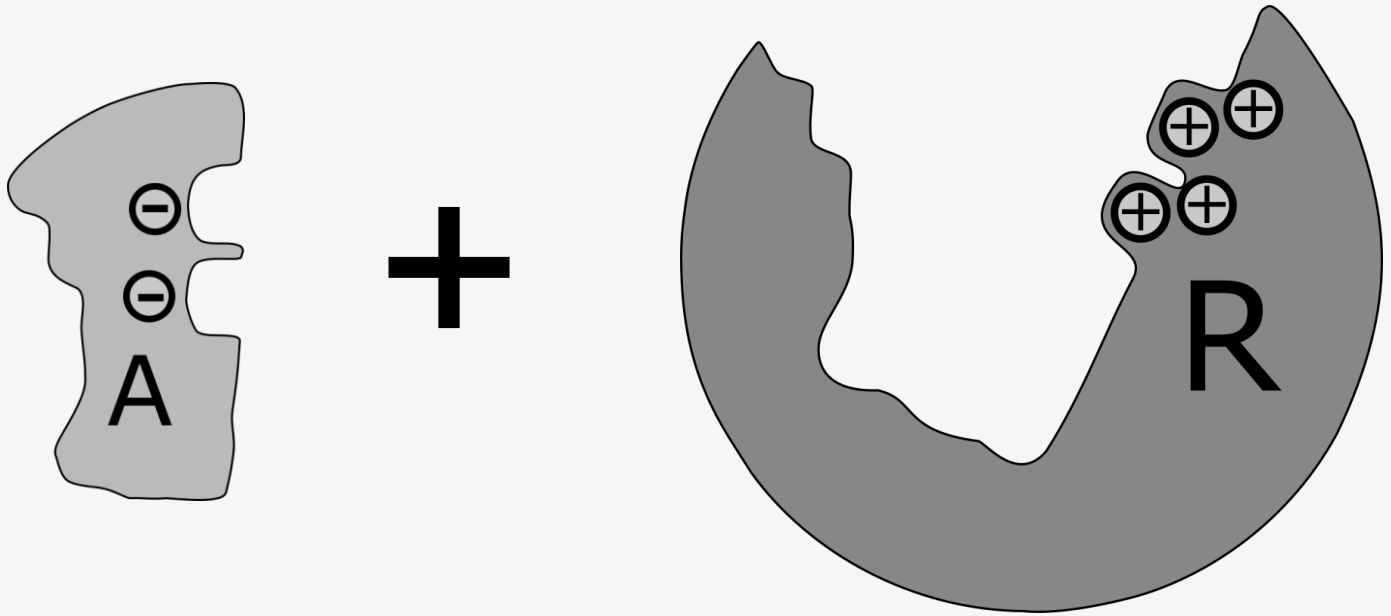


Image from Warne et al., 2011, Nature 469, 241.

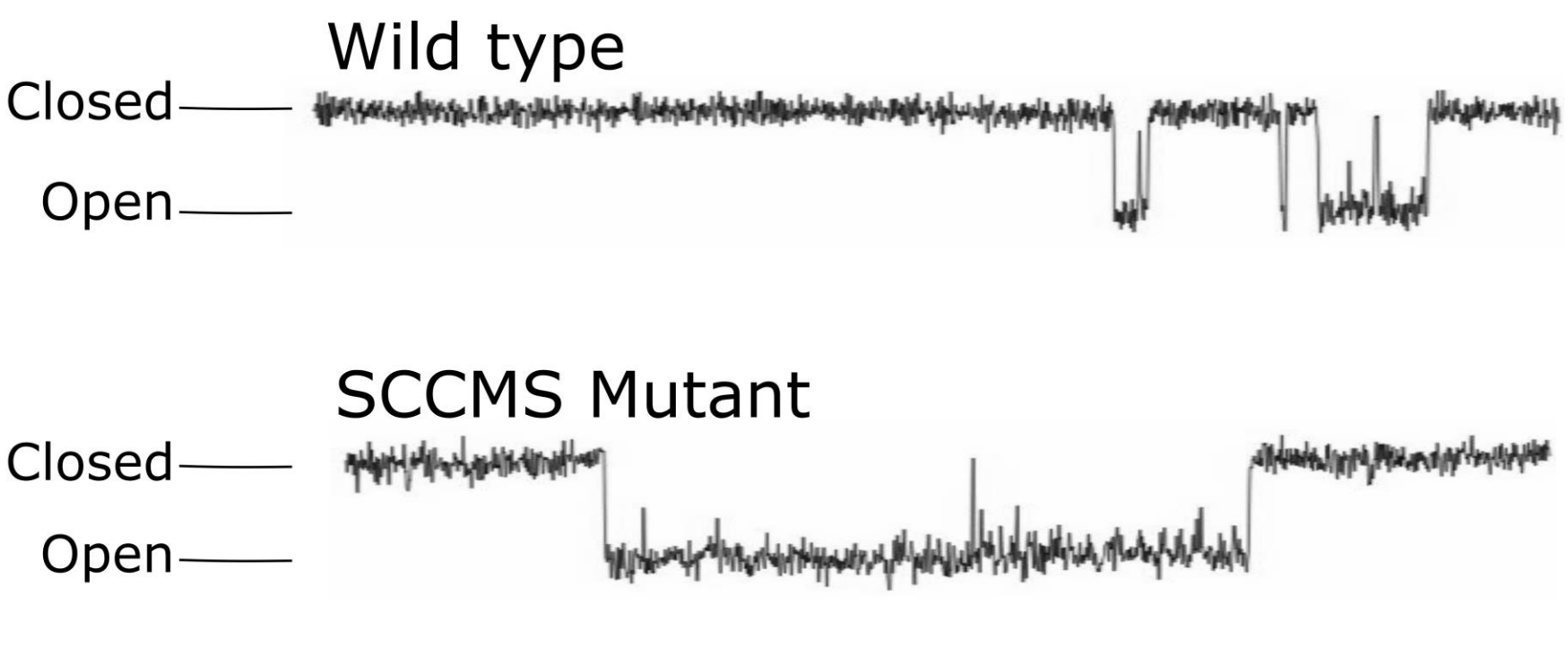
…and we sometimes see something similar for GPCRs.

What if we change the receptor?



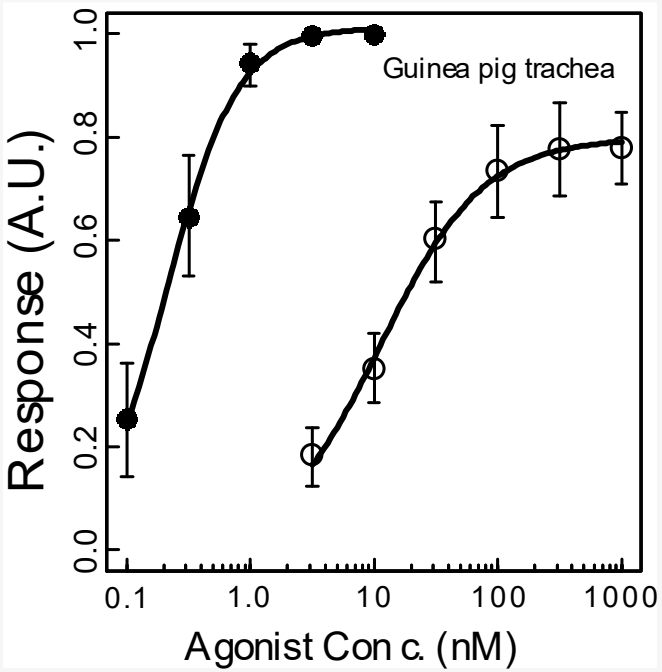
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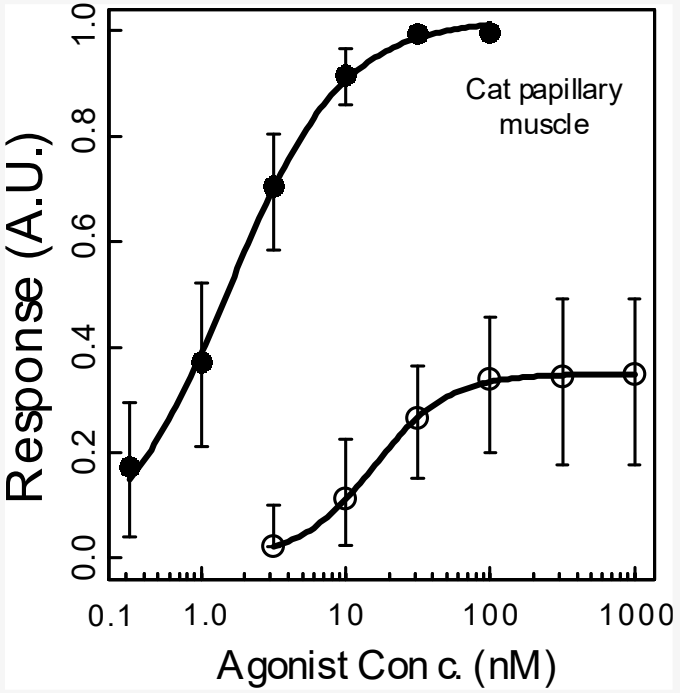
Slow-channel congenital myasthenic syndromes

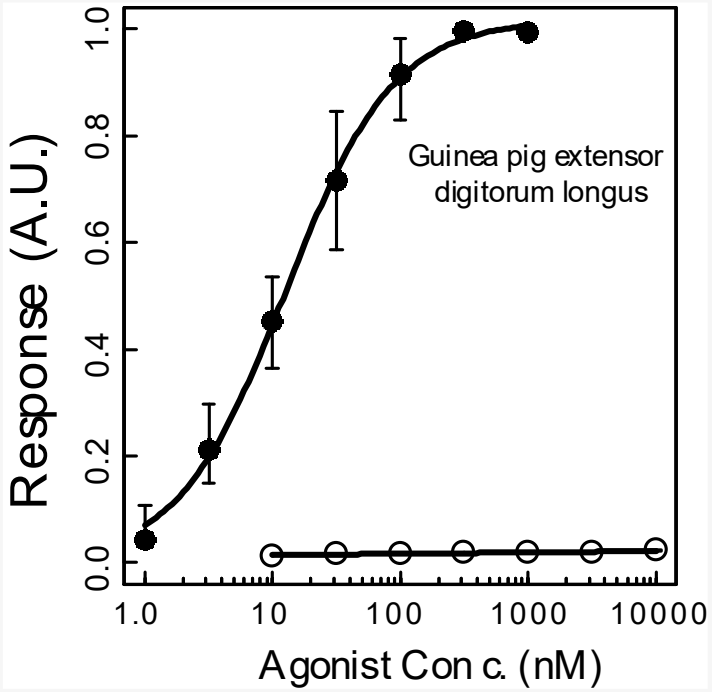


Recordings courtesy of Dr. Chris Shelley UCL, Pharmacology.

Back to some unexplained observations…



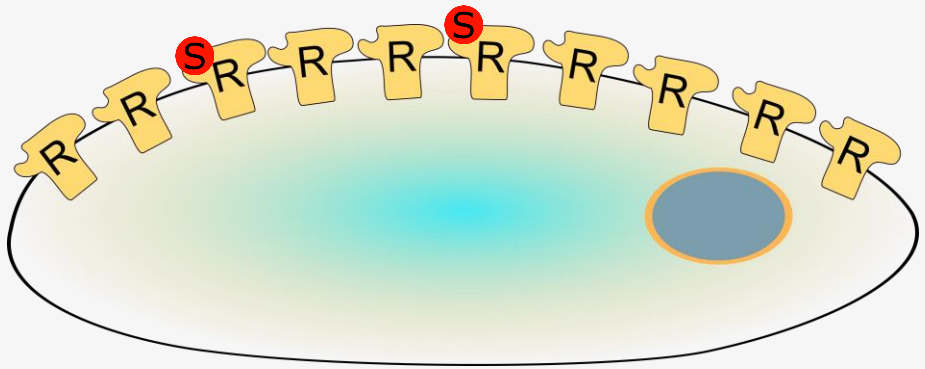




● Isoproteranol
○ Prenalterol

Data reproduced from Kenakin and Beek (1980).

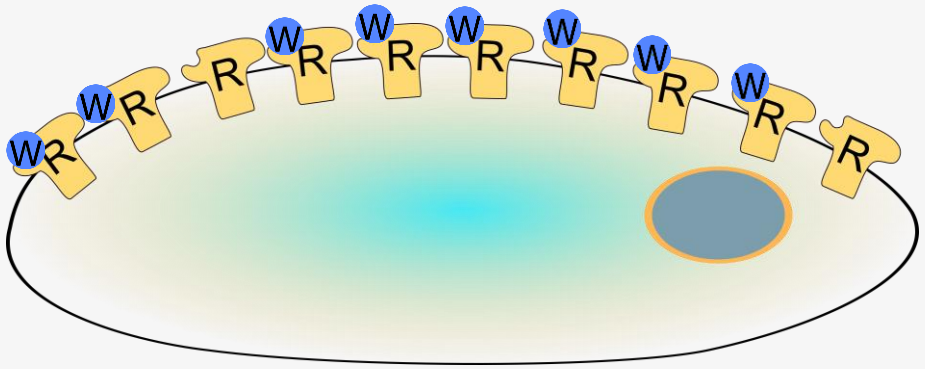
Spare receptors…



For a full agonist each occupied receptors = one unit of activity

2 occupied receptors = 2 units of response

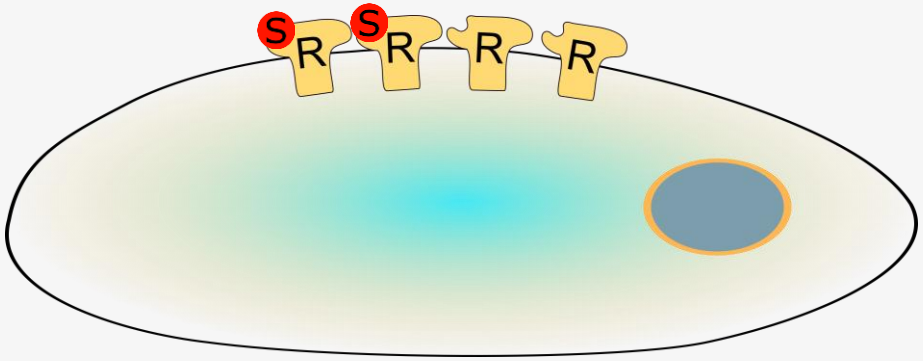
But with a partial agonist…



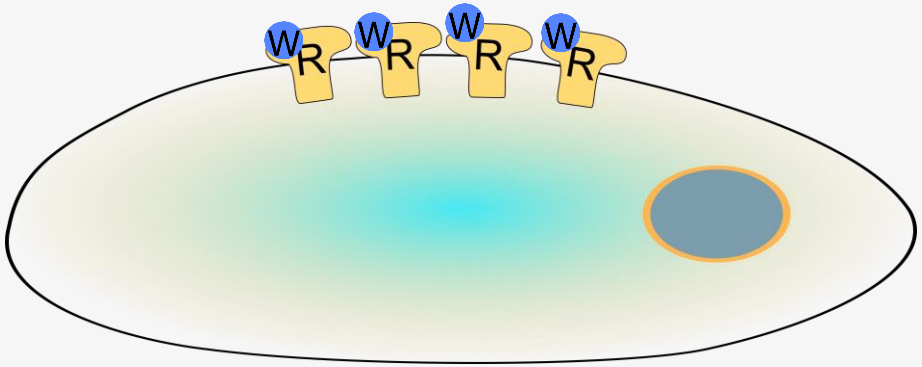
For a partial agonist each occupied receptors = ¼ of a unit of activity

8 occupied receptors = 2 units of response

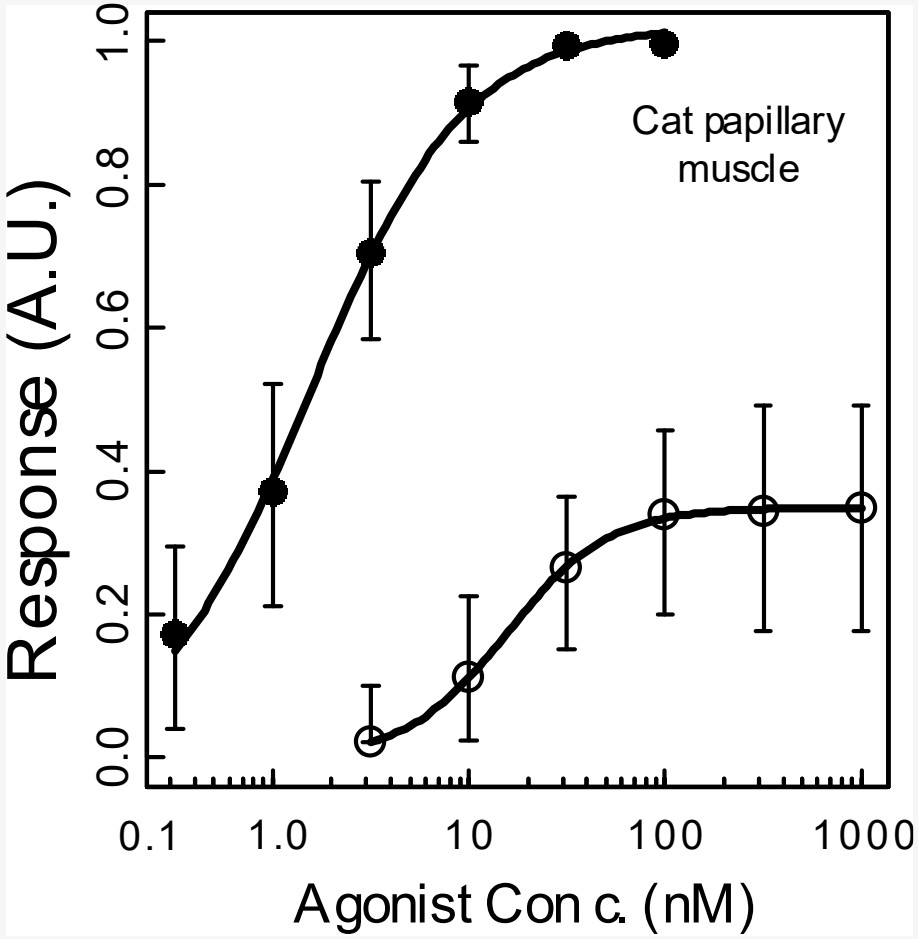
So, if a tissue has fewer spare receptors…



2 occupied receptors = 2 units of response

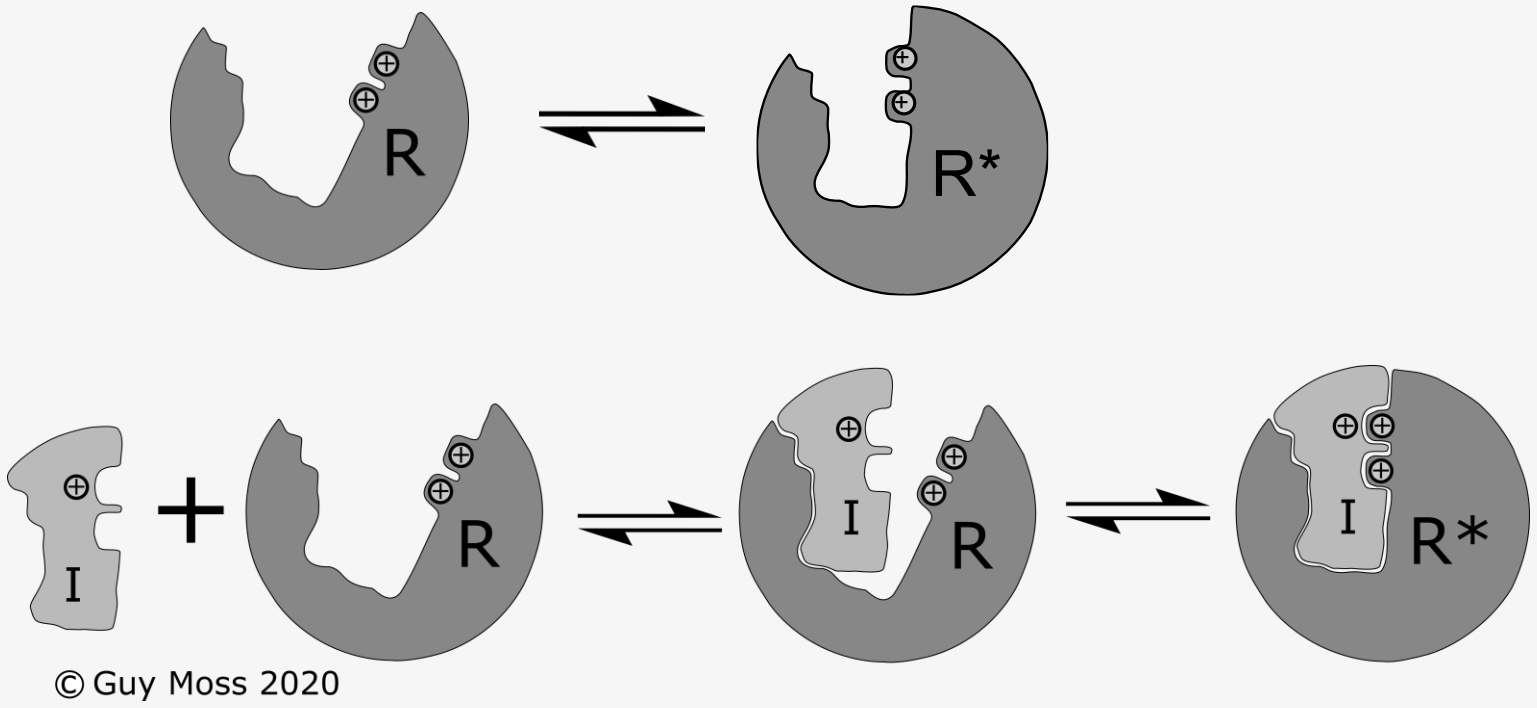


4 occupied receptors = 1 unit of response



Data reproduced from Kenakin and Beek (1980).

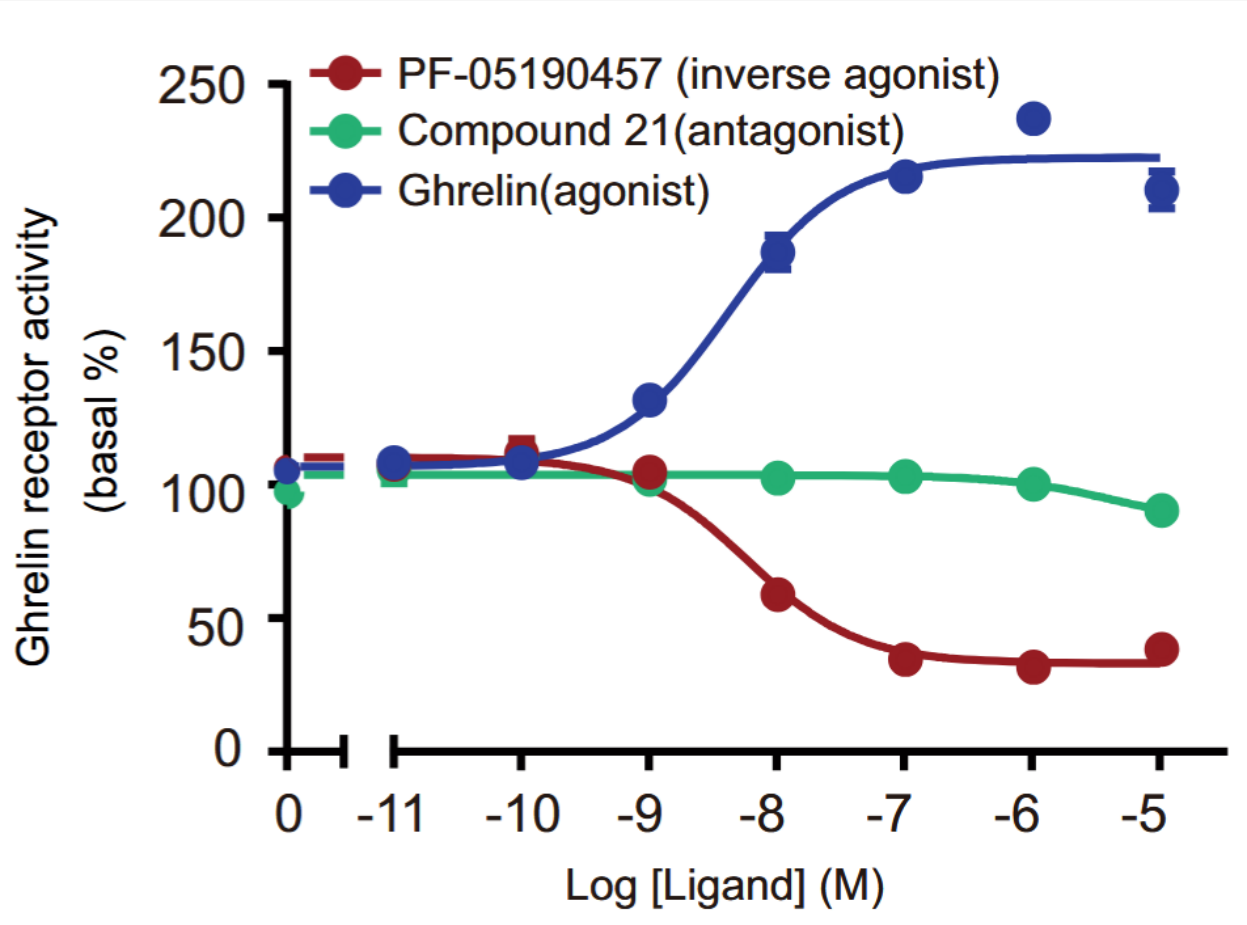
Constitutively active receptors…who needs an agonist?

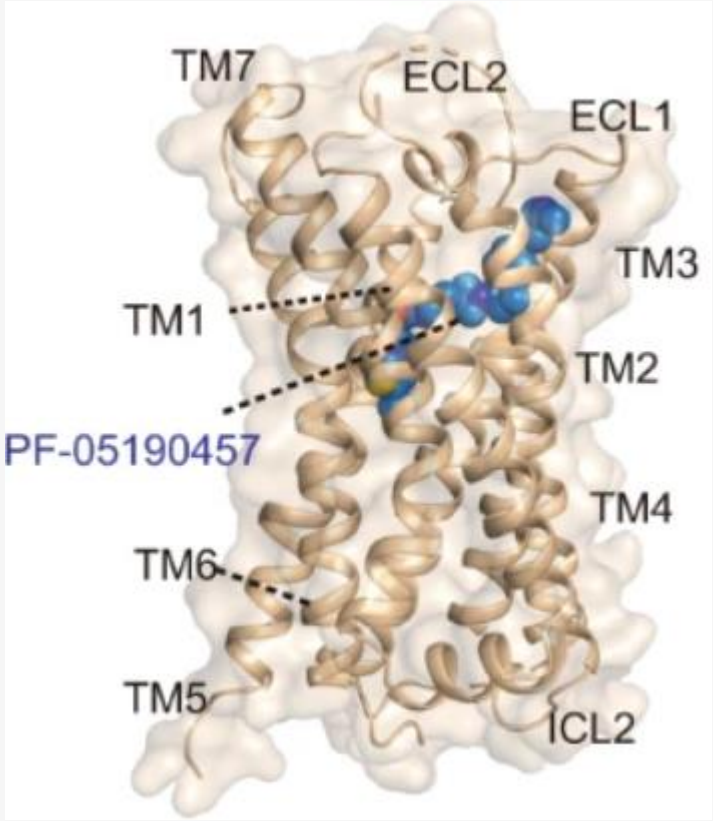


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Inverse agonist

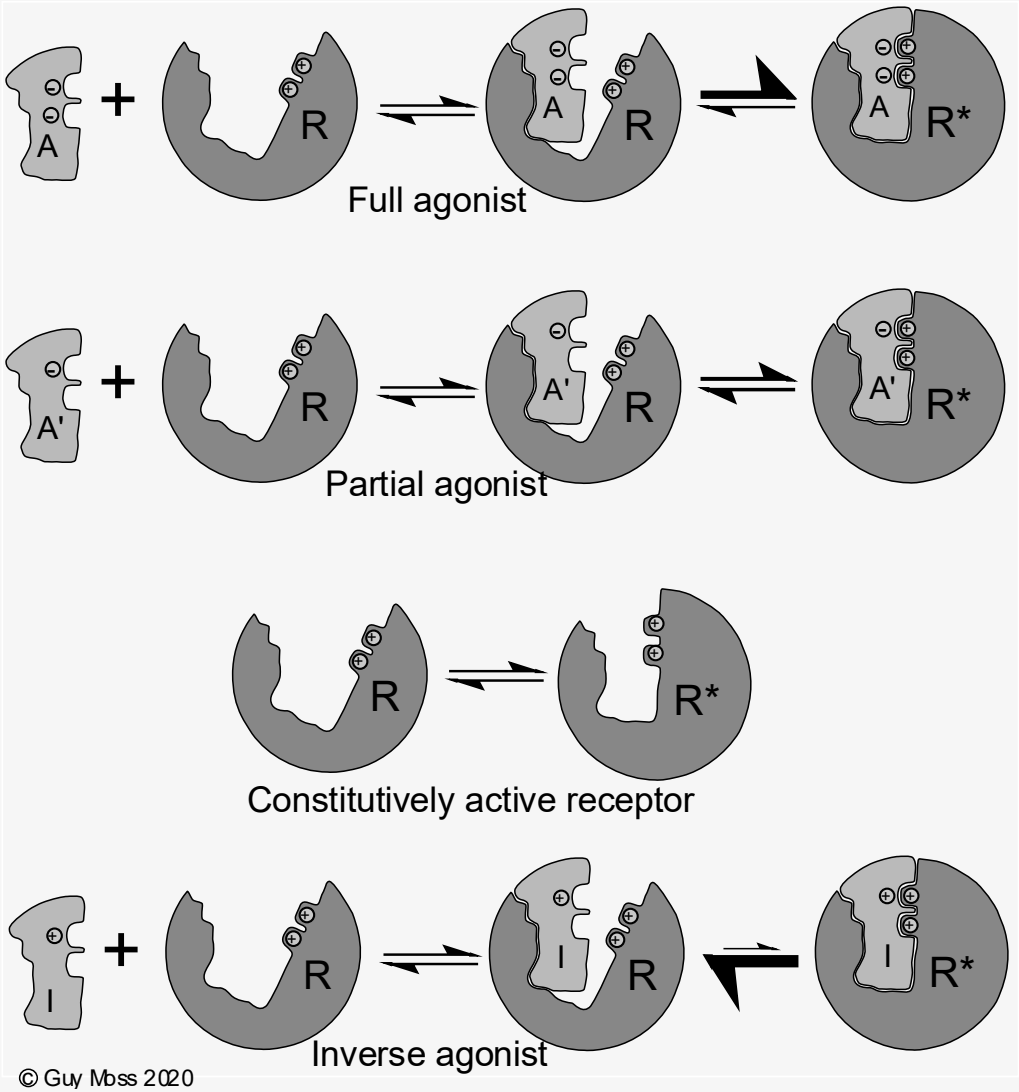
Example, the Ghrelin receptor...





Images from Qin, *et al. Nat Commun* **13**, 300 (2022).

Summary in cartoons…



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Summary

The del Castillo-Katz mechanism:

1. Separates agonist binding and efficacy.
2. Explains partial and inverse agonists.
3. Sheds light on disease mechanisms.
4. Helps us understand spare receptor effects.

Questions…

What is the highest efficacy an agonist can have, and would such a molecule be useful?

Can an artificial agonist ever have a higher efficacy than a natural agonist that produces a full tissue response?